

these checks fail, we abort Droplet creation. If the checks are successful, we proceed to create the Droplet using `CreateDroplet(DropletName, SizeID, OSID, RegID, SshID)`.

The Python function to create a Droplet takes the name, size, OS Image ID, Region ID and the SSH key as arguments, and uses the RESTful API to create the Droplet. A word of caution: the API keys provided here are dummy keys, just for illustrating the flow of the code. You will have to obtain your keys by registering with Digital Ocean.

```
def CreateDroplet (Name, Size, OS, Reg, Passkey):
    #Copying DON's Parameters..
    data = {}

    data['client_id'] = 'xj536XmZsf3NCCznol'
    data['api_key'] = '941c3d1a0240e90ae450848c94'

    data['name'] = Name
    data['size_id'] = Size
    data['image_id'] = OS
    data['region_id'] = Reg
    data['ssh_key_ids'] = Passkey
    URL_Values = urllib.urlencode(data)

    #Connect to DON for values of APIKey...
    URL = 'https://api.digitalocean.com/droplets/new?'
    Full_URL = URL + URL_Values
    print "Droplet Creation URL->[" + Full_URL + "]"
    print "Connecting DON to create droplet"
    print "URL Execution Start..."
    data = urllib2.urlopen(Full_URL)
    DON_Result = data.read()
    print "Droplet Creation Response::[" + DON_Result + "]"
    print "URL Execution End."
    return
```

That's all it takes to create a Droplet. Since we used an SSH key, the root password will not be emailed to us. Log in to the new Droplet using SSH and you'll be prompted for the password since we haven't yet disabled the password authentication in `sshd_conf` configuration. So you'll have to go to the Web console and request for your password or you should not use the SSH keys while creating the Droplet!

Let's now create the users we require and install our infrastructure components—MySQL, PHP, NGINX, Munin, APC, Memcached and Postfix.

Setting up a Droplet

Before installing the components, first set up the time zone, create users, add them to a group and set up the firewall rules. In our case, we set up the time zone to IST, created users, added them to WHEEL (so that they have super cow powers), and then closed all ports except

those we required. You can create this as a shell script called `droplet-admin.bash` or download it from www.opensourcefoxy.com/articles/article_source_code/nov14/cloud_inventory.zip. Run the script to make the above changes or you can do it one by one.

Deploying the cloud stack

Let us now write a script to install PHP Fast CGI, MySQL, Nginx, APC, memcached and Munin.

Let's start with the script for installing the PHP-fCGI. Choose fCGI instead of the conventional PHP module as the former is known to have a lower memory footprint. Create a `php-install.bash` file with the following contents:

```
yum install php php-fpm -y
rpm -ivh http://dl.fedoraproject.org/pub/epel/6/x86_64/epel-release-6-8.noarch.rpm
yum install php-mcrypt -y
yum install php-gd php-imap -y
echo 'cgi.fix_pathinfo=0' >> /etc/php.ini
echo 'date.timezone = America' >> /etc/php.ini
service php-fpm start
service php-fpm status
```

This script installs `php php-fpm`. It then downloads the `php-mcrypt`, `php-gd`, `php-impa` from the `epel` repositories and installs them. `Php-fpm` requires the `cgi.fix_pathinfo=0` to be set in the `php.ini` file, which is done by the `echo` command. The script then automatically starts `php-fpm`.

After PHP, the next thing to be installed is MySQL. Create `mysql-install.bash` by using the following commands:

```
yum install mysql mysql-server -y
chkconfig --levels 235 mysqld on
service mysqld start
service mysqld status
```

The script installs MySQL and configures it to start up automatically when the server starts up. The script currently doesn't remove the demo database. You might want to include that step.

NGINX is not available from the official Centos repositories and the official package has to be downloaded from the NGINX site. Create `nginx-install.bash` with the following lines. This will enable the appropriate repositories and install NGINX:

```
wget http://nginx.org/packages/rhel/6/noarch/RPMS/nginx-release-rhel-6-0.el6ngx.noarch.rpm
rpm -ivh nginx-release-rhel-6-0.el6ngx.noarch.rpm
yum install nginx -y
chkconfig nginx on
service nginx start
service nginx status
```